

Muhammad Isra Al Fattah

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Summary

Informatics student and Full-Stack Software Engineer experienced in building scalable web applications using React, Node.js, and MongoDB. While proficient across the entire stack, core technical strength lies in backend architecture, API design, and system optimization. Building on this engineering rigor with hands-on experience in C# and Unity Engine, and is now channeling these fundamentals into a focused transition toward Game Development. Currently expanding practical software engineering capabilities as a Research Software Engineer Intern at GAW Bukit Kototabang Station and a Full-Stack Developer Cohort Member at DBS Foundation's Coding Camp 2026.

Experience

Global Atmosphere Watch (GAW) Station

Research Software Engineer (Intern)

Bukittinggi, West Sumatra

June 2026 – Present

- Designing, maintaining, and developing software solutions to support atmospheric monitoring operations.
- Facilitating interactive data visualization systems and supporting ongoing meteorological research initiatives.

DBS Foundation's Coding Camp 2026 (Dicoding)

Full-Stack Developer Cohort

Remote

February 2026 – June 2026

- Architected a scalable backend infrastructure for the HEARTZ speech therapy MVP, integrating 2 core AWS cloud services (RDS PostgreSQL and S3) to securely manage relational data and audio object storage.
- Engineered a parallel proxy gateway that seamlessly bridged frontend React audio streams to 2 deep learning AI models (2D CNN and Mel-Spectrogram) for real-time processing.
- Developed secure RESTful APIs utilizing JSON Web Tokens (JWT) and HTTP-Only cookies, collaborating via Git/GitHub to successfully deliver the project within a strict 1-month Capstone timeline.
- Completed an intensive full-stack web development curriculum across 10 structured technical modules, mastering dynamic frontend UI building with React.js and robust server-side engineering with Node.js.

Projects

Heartz

AI-powered speech therapy web application

React, Node.js, Express.js

PostgreSQL, AWS S3, FastAPI

- Built a responsive web application utilizing React.js for the frontend interface and Node.js/Express.js for the backend server architecture.
- Engineered a high-performance proxy gateway handling multi-format audio uploads via Multer to stream data into 2 deep learning AI models concurrently.
- Integrated cloud infrastructure by provisioning AWS RDS (PostgreSQL) for user data persistence and AWS S3 for secure, remote audio object storage.
- Fortified application security with JWT authentication and HTTP-Only cookies, utilizing Git/GitHub to deploy a stable MVP within a 1-month timeline.

Ruang Belajar

Flutter, Google ML Kit, SQLite

On-device AI mobile productivity application

- Built a mobile application to track study concentration utilizing front-camera face detection via Google ML Kit.
- Implemented automated daily task calendar scheduling powered by the Gemini API and stored productivity score data locally using SQLite.

Astro View

Node.js, Express,

Real-time NASA data aggregator web application

MongoDB Atlas, Socket.IO

- Developed a microservice data processing web application utilizing NASA APIs, monitoring visitor traffic in real-time via WebSocket.
- Integrated an automated daily newsletter delivery system and implemented geospatial data filtering from NASA EONET.

HeadGear_Classification

Flutter, TensorFlow Lite, Native C++

Offline edge AI image classification mobile application

- Designed a serverless mobile application with Flutter to run computer vision models directly on edge devices.
- Integrated a TensorFlow Lite (TFLite) classifier to identify 20 distinct classes of headgear completely offline with zero network latency.
- Formulated custom input normalization, filtered Top-K predictions with a >20% confidence threshold, and enforced a strict memory-cleanup pipeline to prevent native C++ memory leaks.

Skills & Interests

Technical: C++, C#, Unity, Javascript, Node.js, React.js, PostgreSQL, Mongo DB, Next.js, Express.js, HTML5, CSS3, Bootstrap, Git, Firebase, Docker, Python, Flutter, Tailwind, Postman, AWS

Language: English (Professional Working Proficiency), Indonesian (Native)

Interests: Single-player video games, playing guitar, fitness, reading fiction and non-fiction, watching movies, and listening to music.

Education

Universitas Negeri Padang

West Sumatra, Indonesia

Bachelor of Informatics

Expected Graduation : 2027

GPA : 3.67 / 4.00